

Soft Voting Ensemble Learning and SHAP-Based Interpretability for Maternal and Fetal Health Risk Stratification Across Multiple Clinical Datasets

1st Dorothy Shahnaz Mukul

*Department of Obstetrics and Gynecology
Shahid Syed Nazrul Islam Medical College
Kishoreganj-2300, Bangladesh
fatema89@gmail.com*

2nd Pratima Rani Bishas

*Department of Obstetrics and Gynecology
Manikganj Medical College Hospital
Manikganj-1800, Bangladesh
medisunhealthcare@gmail.com*

3rd Azizur Rahman

*Department of CAPS (Computer Technology)
Indiana Wesleyan University
Indiana-46953, United States of America
azizurusa22@gmail.com*

4th Nakib Uddin

*Department of Computer Science and Engineering
Metropolitan University
Sylhet-3101, Bangladesh
nakibuddin33@gmail.com*

5th Md Shahin Shah

*Department of Computer Science and Engineering
Port City International University
Chittagong-4225, Bangladesh
shahin.ai.engineer.uk@gmail.com*

6th Shamim MD Jony

*Department of Computer Science and Engineering
Port City International University
Chittagong-4225, Bangladesh
shamimjonym@gmail.com*

Abstract—Maternal and fetal health risk assessment is a vital issue in clinical practice, and early, accurate identification of high-risk pregnancies can significantly reduce maternal and neonatal mortality. This paper presents a machine learning framework for automated risk classification evaluated on three benchmark datasets: a binary maternal high-risk pregnancy dataset (D1, $n = 1,205$), a multi-class maternal health risk dataset (D2, $n = 1,014$), and a fetal health classification dataset (D3, $n = 2,126$). Six classifiers (Random Forest, Support Vector Machine, K-Nearest Neighbors, XGBoost, Decision Tree, and a Soft Voting Ensemble) were evaluated with 5-fold stratified cross-validation. The Soft Voting Ensemble achieved the highest mean accuracy on D1 (98.99%) and D2 (86.20%), while XGBoost achieved the best performance on D3 (95.39%). The Friedman test showed statistically significant inter-model differences on all three datasets ($p < 0.005$), and post-hoc Wilcoxon signed-rank tests indicated a consistent, though not always statistically significant, advantage of the Ensemble over the baselines. SHAP-based explainability applied to the best-performing model on each dataset produced clinically interpretable feature attributions, highlighting blood glucose, systolic blood pressure, and abnormal short-term variability as dominant predictors. These results indicate that combining ensemble learning, rigorous statistical validation, and explainability offers a transparent and reliable platform to support clinical decisions in maternal and fetal healthcare.

Index Terms—explainable artificial intelligence (XAI), fetal health classification, high-risk pregnancies, maternal health risk, Wilcoxon signed-rank test.

I. INTRODUCTION

Complications during pregnancy remain a leading contributor to maternal and neonatal morbidity and mortality worldwide, and early identification of high-risk pregnancies is essential for timely clinical intervention [1]. However, the nonlinear interactions among physiological, clinical, and demographic determinants make accurate risk prediction inherently difficult, and conventional statistical approaches such as logistic regression often fail to capture these complex relationships [1], [2]. Machine learning (ML) offers a data-driven alternative capable of automatically modeling nonlinear dependencies in clinical data [3], but most existing studies report only headline accuracy from a single train-test split, without assessing model stability, statistical significance, or interpretability – all of which are prerequisites for clinical trust and adoption.

To address these gaps, this work proposes a multi-dataset ML framework for maternal and fetal health risk classification with the following contributions:

- A comparative evaluation of six classifiers (RF, SVM, KNN, XGBoost, DT, and a Soft Voting Ensemble) across three clinically distinct benchmark datasets using 5-fold stratified cross-validation, with per-fold stability reporting.
- Formal statistical validation of inter-model differences using the Friedman test and pairwise Wilcoxon signed-

rank post-hoc analysis.

- A SHAP-based explainability analysis of the best-performing model on each dataset, providing global and patient-level feature attributions for clinical interpretability.

II. LITERATURE REVIEW

Machine learning has shown strong potential for maternal and fetal risk prediction. Tree-based and ensemble models trained on clinical history and physiological features have demonstrated competitive predictive performance for pregnancy complications and fetal abnormalities [3], [4], with random forest consistently identified as a strong baseline for maternal mortality and high-risk classification, including in IoT-based monitoring settings [5], [6]. Building on these baselines, ensemble methods – particularly gradient boosting and multi-learner architectures – have outperformed traditional regression for outcomes such as preterm birth and gestational diabetes [2], [7], and have proven especially useful for sparse or noisy data in low- and middle-income settings [8]. Soft voting ensembles that combine calibrated class probabilities have been shown to sharpen decision boundaries and improve maternal risk stratification [9], and gradient-boosted pipelines combined with resampling or dimensionality reduction have reported strong results across independent obstetric studies [10]–[12], although some population-based cohorts show only marginal gains over conventional models, suggesting that predictive success is highly sensitive to data quality and feature engineering [13].

In parallel, Explainable Artificial Intelligence (XAI) has become essential for clinical decision support, since the black-box nature of high-performing ML models limits their adoption unless predictions are accompanied by interpretable rationale [14], [15]. SHAP, in particular, has been used to visualize feature contributions in hybrid fuzzy-gradient-boosting models [16] and to identify socio-demographic determinants of maternal care outcomes [17], consistently highlighting physiologically plausible predictors such as blood glucose, blood pressure, and pre-existing diabetes [14], [18], while hybrid deep-learning architectures have further improved accuracy at the cost of interpretability [19]. Despite this progress, key limitations persist: many studies rely on small or region-specific datasets that limit generalizability [20], class imbalance continues to hinder classification of intermediate-risk groups [7], interpretable and clinically transparent models remain scarce [15], and cross-dataset generalizability within a single, statistically validated framework has rarely been assessed – the gap this work aims to close.

III. METHODOLOGY

This section describes the reproducible machine learning pipeline used for maternal and fetal health risk classification, covering dataset description, preprocessing, model training, cross-validation, statistical testing, and SHAP-based explainability. Fig. 1 shows the end-to-end workflow.

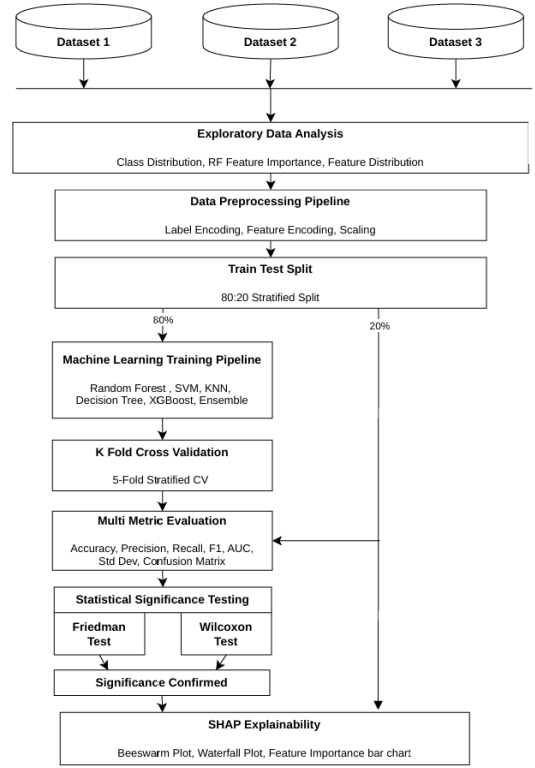


Fig. 1. End-to-end methodological pipeline for maternal and fetal health risk classification.

A. Dataset Description

Three publicly available benchmark datasets were used to evaluate the proposed framework across tasks of varying complexity and clinical scope: D1 for binary high-risk pregnancy classification, D2 for three-level maternal health risk stratification, and D3 for fetal health status classification from cardiotocography (CTG) measurements. Table I summarizes their characteristics.

TABLE I
CHARACTERISTICS OF THE THREE BENCHMARK DATASETS.

ID	Dataset	Rec.	Task	Classes
D1	Maternal High-Risk	1,205	Binary	High/Low [21]
D2	Maternal Health Risk	1,014	Multi-class	Low/Mid/High [22]
D3	Fetal Health	2,126	Multi-class	Normal/Suspect/Path. [23]

B. Exploratory Data Analysis and Preprocessing

Histogram and kernel density estimation (KDE) plots were used to inspect feature distributions, skewness, and outliers, and class imbalance was quantified to guide stratification and metric choice: D1 is moderately imbalanced (high-risk: 40.8%), D2 is fairly balanced, and D3 is highly imbalanced (pathological class: 8.3%). Preprocessing followed a standardized three-stage pipeline. Target labels were integer-encoded,

with D2 using ordinal encoding to preserve the natural risk hierarchy. Categorical (object-type) features were encoded with scikit-learn’s `LabelEncoder` prior to scaling. Numerical features were standardized (zero mean, unit variance) using `StandardScaler`, applied within cross-validation folds for SVM and KNN to prevent data leakage, and globally for scale-invariant tree-based models (RF, XGBoost, DT). All datasets were split 80:20 into stratified training and test subsets; the test set was reserved exclusively for SHAP analysis, while model training and evaluation used cross-validation.

C. Classification Models and Hyperparameters

Six supervised classifiers spanning linear, instance-based, tree-based, gradient-boosted, and ensemble paradigms were evaluated (Table II). The Soft Voting Ensemble combined the predicted class probabilities of SVM, KNN, DT, and XGBoost with voting weights [1, 1, 1, 2], giving XGBoost greater weight due to its superior probability calibration in preliminary experiments. Random Forest was deliberately excluded from the ensemble and retained as an independent baseline to assess the ensemble’s incremental value over the strongest single model.

TABLE II
HYPERPARAMETER SETTINGS AND IMPLEMENTATION DETAILS.

Model	Key Hyperparameters	Implementation
RF	n_estimators=300; max_depth=None; criterion=gini	Trained independently per fold
SVM	kernel=rbf; C=1.0; gamma=scale; probability=True	Pipeline with StandardScaler (in-fold)
KNN	n_neighbors=5; weights=distance; metric=euclidean	Pipeline with StandardScaler (in-fold)
XGBoost	n_estimators=500; lr=0.05; max_depth=6; subsample=0.8; colsample=0.8; reg_lambda=1.0	L2 regularization; row/column subsampling
DT	criterion=gini; max_depth=None; min_samples_split=2	Unpruned, interpretable baseline
Ensemble	Base: SVM, KNN, DT, XGBoost; voting=soft; weights=[1,1,1,2]	RF excluded, kept as standalone baseline

D. Training, Evaluation, and Statistical Testing Protocol

Each classifier was evaluated using 5-fold Stratified K-Fold cross-validation, preserving class distribution across folds – particularly important given the imbalance in D1 and D3. Performance was reported as mean accuracy, standard deviation, macro-averaged precision/recall/F1-score, and macro-AUC-ROC across folds. Inter-model differences were tested globally with the non-parametric Friedman test and, where significant, followed by pairwise Wilcoxon signed-rank tests comparing the Ensemble against each baseline. Finally, SHAP (SHapley Additive exPlanations) was applied to the best-performing model on each dataset, retrained on the 80% training split and evaluated on up to 200 stratified test instances, to produce

global (beeswarm, mean |SHAP|) and patient-level (waterfall) feature attributions.

IV. RESULTS AND DISCUSSION

A. Cross-Validation Performance

1) *D1 – Binary Maternal High-Risk Classification:* As shown in Table III, the Ensemble achieved the highest mean accuracy (98.99%, SD = 0.43), narrowly ahead of Random Forest (98.90%) and XGBoost (98.74%). The Decision Tree was the least accurate (96.80%) and most variable (SD = 1.59), reflecting its sensitivity to variance without pruning. XGBoost attained the best macro precision, recall, and F1-score (all 1.00) with AUC = 0.9994. All models exceeded AUC = 0.96, but the Decision Tree’s higher variance warrants caution for clinical deployment.

2) *D2 – Multi-Class Maternal Health Risk Stratification:* Table IV shows the Ensemble achieving the highest mean accuracy (86.20%, SD = 2.23) and macro-AUC (0.9578), closely followed by XGBoost (85.80%, AUC = 0.9557). The relatively high standard deviations across all models (2.13–2.90) reflect the difficulty of separating the Mid-Risk class. RF and XGBoost both exceeded macro-AUC 0.95, and the Ensemble’s modest gain over XGBoost (+0.40%) illustrates the limited but consistent benefit of soft voting when the best single learner already performs well.

3) *D3 – Multi-Class Fetal Health Classification:* For D3 (Table V), XGBoost achieved the highest mean accuracy (95.39%, SD = 0.91) and AUC (0.9889), marginally ahead of the Ensemble (94.92%, SD = 0.73) and Random Forest (94.78%, AUC = 0.9883) – the only dataset where the Ensemble is not ranked first. The Decision Tree again underperformed, consistent with its known bias toward majority classes absent explicit balancing, while the high AUC of XGBoost and RF confirms strong discriminative power despite severe class imbalance.

B. Comparative Model Analysis

Across all three datasets, the Soft Voting Ensemble and XGBoost consistently ranked first and second, confirming the effectiveness of ensemble and gradient-boosted methods on structured clinical tabular data, while Random Forest remained a consistently strong third. SVM was the most task-sensitive model, competitive on binary D1 but degrading substantially on the multi-class D2 task. Fig. 2 and Fig. 3 present the confusion matrices and ROC curves for the top-performing model on each dataset.

C. Statistical Significance Testing

The Friedman test (Table VI) confirmed statistically significant differences among the six classifiers on all three datasets ($p < 0.005$), rejecting the null hypothesis of equal performance and justifying the pairwise post-hoc analysis that follows.

Table VII summarizes pairwise Wilcoxon signed-rank tests between the Ensemble and each baseline. With only five folds, the smallest achievable two-tailed p-value is 0.0625, so no comparison reached formal significance; nonetheless, the

TABLE III
CROSS-VALIDATION PERFORMANCE SUMMARY FOR DATASET D1.

Model	Mean Acc. (%)	Std Dev (%)	Precision	Recall	F1-Score	AUC (Macro)	Fold Accuracies (%)
Random Forest	98.90	0.43	0.99	0.99	0.99	0.9994	98.74 / 99.16 / 99.58 / 98.73 / 98.31
SVM	97.39	0.62	0.98	0.98	0.98	0.9980	96.64 / 98.32 / 97.05 / 97.05 / 97.89
KNN	97.30	0.57	0.98	0.98	0.98	0.9923	96.22 / 97.90 / 97.47 / 97.47 / 97.47
XGBoost	98.74	0.60	1.00	1.00	1.00	0.9994	99.16 / 98.74 / 99.58 / 98.31 / 97.89
Decision Tree	96.80	1.59	0.99	0.99	0.99	0.9659	97.48 / 98.74 / 94.51 / 95.36 / 97.89
Ensemble	98.99	0.43	0.99	0.99	0.99	0.9992	99.16 / 99.16 / 99.58 / 98.73 / 98.31

TABLE IV
CROSS-VALIDATION PERFORMANCE SUMMARY FOR DATASET D2.

Model	Mean Acc. (%)	Std Dev (%)	Precision	Recall	F1-Score	AUC (Macro)	Fold Accuracies (%)
Random Forest	85.21	2.27	0.89	0.89	0.89	0.9563	87.19 / 83.74 / 83.74 / 82.76 / 88.61
SVM	70.02	2.90	0.74	0.74	0.73	0.8360	70.94 / 70.94 / 65.02 / 73.89 / 69.31
KNN	82.94	2.13	0.86	0.85	0.85	0.9368	81.28 / 85.22 / 83.25 / 79.80 / 85.15
XGBoost	85.80	2.68	0.90	0.90	0.90	0.9557	89.66 / 86.70 / 82.76 / 82.76 / 87.13
Decision Tree	84.72	2.89	0.89	0.88	0.88	0.9324	86.70 / 86.21 / 81.77 / 80.79 / 88.12
Ensemble	86.20	2.23	0.90	0.89	0.89	0.9578	88.18 / 86.21 / 84.24 / 83.25 / 89.11

TABLE V
CROSS-VALIDATION PERFORMANCE SUMMARY FOR DATASET D3.

Model	Mean Acc. (%)	Std Dev (%)	Precision	Recall	F1-Score	AUC (Macro)	Fold Accuracies (%)
Random Forest	94.78	0.81	0.96	0.96	0.96	0.9883	95.77 / 93.65 / 94.82 / 94.12 / 95.53
SVM	90.73	1.20	0.93	0.93	0.93	0.9745	91.31 / 92.71 / 89.41 / 90.59 / 89.65
KNN	91.06	0.72	0.92	0.92	0.92	0.9567	91.55 / 90.82 / 89.88 / 91.06 / 92.00
XGBoost	95.39	0.91	0.96	0.96	0.96	0.9889	96.24 / 93.88 / 96.24 / 94.82 / 95.76
Decision Tree	91.25	0.82	0.92	0.92	0.92	0.8915	91.55 / 89.65 / 91.53 / 92.00 / 91.53
Ensemble	94.92	0.73	0.96	0.96	0.96	0.9874	96.24 / 94.12 / 95.06 / 94.59 / 94.59

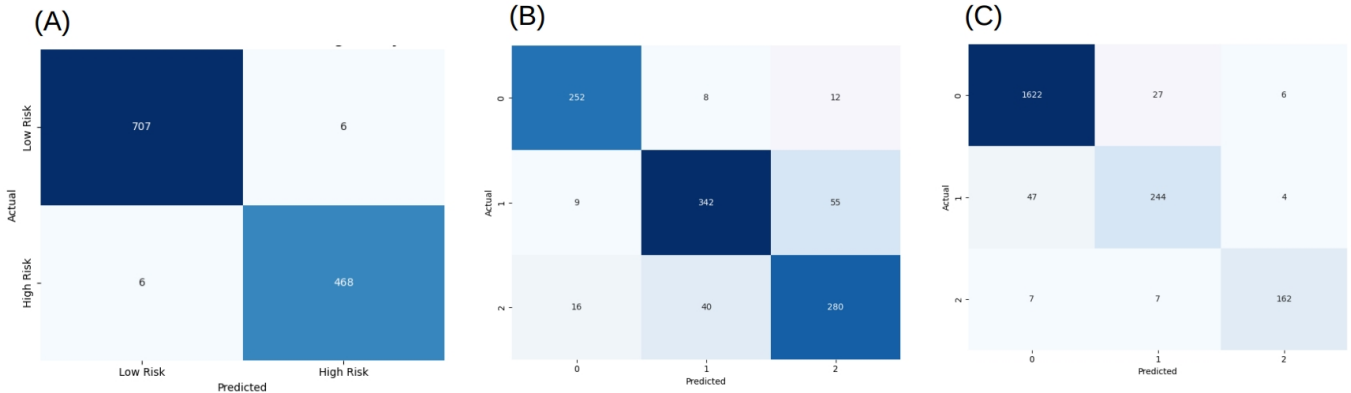


Fig. 2. Confusion matrices for the top-performing model on each dataset: (A) Ensemble on D1, (B) Ensemble on D2, and (C) XGBoost on D3.

TABLE VI
FRIEDMAN TEST RESULTS (5-FOLD ACCURACY DISTRIBUTIONS).

Dataset	χ^2	p-value	Significant ($\alpha=0.05$)
D1	19.5679	0.0015	Yes
D2	18.0058	0.0029	Yes
D3	21.9364	0.0005	Yes

directional evidence consistently favors the Ensemble, which achieved the best or joint-best accuracy in 8 of 15 model–dataset combinations and a positive mean-accuracy advantage over four of five baselines on every dataset. The sole reversal, Ensemble vs. XGBoost on D3 ($\delta = -0.47$), is clinically

negligible.

TABLE VII
WILCOXON SIGNED-RANK TEST: ENSEMBLE VS. BASELINES (MIN. ACHIEVABLE $p = 0.0625$ at $n=5$ FOLDS).

Baseline	D1 δ (Acc.%)	D2 δ	D3 δ
Random Forest	+0.09	+0.99	+0.14
SVM	+1.60	+16.18	+4.19
KNN	+1.69	+3.26	+3.86
XGBoost	+0.25	+0.40	−0.47
Decision Tree	+2.19	+1.48	+3.67

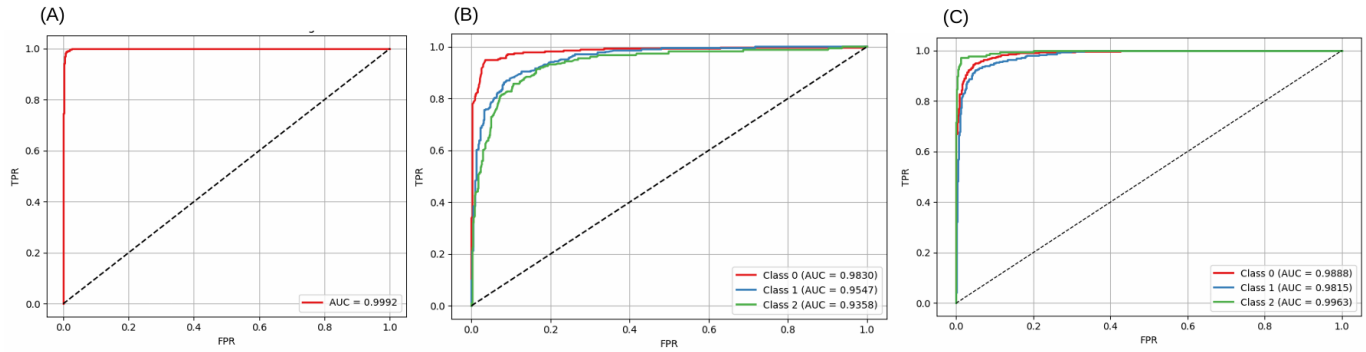


Fig. 3. ROC curves for the top-performing model on each dataset: (A) Ensemble on D1, (B) Ensemble on D2, and (C) XGBoost on D3.

D. Explainability Analysis via SHAP

SHAP was applied to the best-performing model on each dataset (Ensemble on D1/D2, XGBoost on D3) to obtain global and patient-level feature attributions. On D1, blood glucose and systolic blood pressure emerged as the dominant predictors of high-risk pregnancy, with body temperature and heart rate contributing secondary, smaller effects; a representative waterfall case showed elevated glucose and blood pressure jointly driving a high-risk prediction, partially offset by a moderately low heart rate. On D2, the same two physiological features remained influential, but their SHAP distributions were more dispersed, consistent with the added difficulty of separating three risk classes and the frequent Low/Mid-Risk misclassifications observed in the confusion matrix; age and body temperature contributed secondary, more variable effects. On D3, the CTG-derived features abnormal short-term variability (ASTV) and mean short-term variability (MSTV) were the most influential, together with accelerations and prolonged decelerations – a pattern consistent with established clinical CTG interpretation, where absent accelerations and late decelerations signal fetal compromise. Across all three datasets, the SHAP attributions aligned with recognized clinical risk factors, supporting the physiological plausibility of the learned models and their suitability as transparent clinical decision-support tools.

V. CONCLUSION

This work presented a machine learning framework for maternal and fetal health risk classification, evaluating six classifiers on three benchmark datasets with 5-fold stratified cross-validation. The Soft Voting Ensemble achieved the best mean accuracy on D1 (98.99%) and D2 (86.20%), while XGBoost performed best on D3 (95.39%). The Friedman test confirmed statistically significant inter-model differences ($p < 0.005$), and SHAP analysis produced clinically plausible feature attributions – blood glucose and systolic blood pressure for maternal risk, and abnormal short-term variability for fetal health classification. Limitations include reliance on benchmark datasets and the limited statistical power of a five-fold design. Future work will explore deep learning architectures,

fairness-aware modeling, multi-modal data integration, and prospective clinical validation.

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